MATHEMATICS

Maximum Marks: 80 **Time Allowed: Three Hours**

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed an additional **fifteen minutes** for **only** reading the paper.
2. You must **NOT** start writing during reading time.
3. The question paper has **10** printed pages.
4. It consists of **twenty** questions and **four** sections: Sections A, B, C and D.
5. All questions are compulsory.
6. Section A consists of very short answer questions of 1 mark each.
7. Section B consists of short answer questions of 2 marks each.
8. Section C consists of moderately long answer questions of 3 marks each.
9. Section D consists of long answer questions of 5 marks each.
10. Internal choices have been provided in **three** questions, each in Sections B, C and D.
11. While attempting Multiple Choice Questions in Section A, you are required to write only **ONE** option as the answer.
12. The intended marks for questions or parts of questions are given in the brackets [ ].
13. All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
14. Mathematical tables and graph papers are provided.

Instruction to Supervising Examiner

Kindly read aloud the instructions given above to all the candidates present in the examination hall.

SECTION A – 20 MARKS

Question 1

In subparts (i) to (xvii) choose the correct options and in subparts (xviii) to (xx), answer the questions as instructed.

(i) Let L be the set of all straight lines in a plane. The relation R on L defined by L_1 R L_2 if and only if L_1 and L_2 intersect (have at least one point in common) is: *(Understand)* [1]

(a) an equivalence relation

(b) symmetric and transitive but not reflexive

(c) reflexive and symmetric but not transitive

(d) reflexive and transitive but not symmetric

(ii) The value of $\cot\left(\frac{\pi}{2} + \tan^{-1}2\right)$ is: *(Apply)* [1]

(a) 2

(b) $\frac{1}{2}$

(c) $-\frac{1}{2}$

(d) −2

(iii) If $A = \begin{bmatrix} 3 & 1 \\ 5 & 4 \end{bmatrix}$, then |A (adj A)| is equal to: *(Apply)* [1]

(a) 49

(b) 7

(c) 14

(d) 343

(iv) In which interval is the function $f(x) = x^3 - 27x$ increasing? *(Apply)* [1]

(a) (−3, 3)

(b) $(-\infty, -3) \cup (3, \infty)$

(c) $(-3, \infty)$

(d) $(-\infty, 3)$

(v) **Statement I:** $f(x) = 5 + 2x - e^x$ is strictly increasing in $(-\infty, 0)$. [1]

Statement II: $f(x) = 5 + 2x - e^x$ is strictly decreasing in $(0, \infty)$.

(Understand)

(a) Statement I is true and Statement II is false.

(b) Statement I is false and Statement II is true.

(c) Both the statements are true.

(d) Both the statements are false.

(vi) If $y = \log\sqrt{\sin x}$, then $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$ is: *(Apply)* [1]

(a) ∞

(b) 1

(c) 0

(d) $\frac{1}{2}$

(vii) The value of $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{1 + x^2}\, dx$ is: *(Apply)* [1]

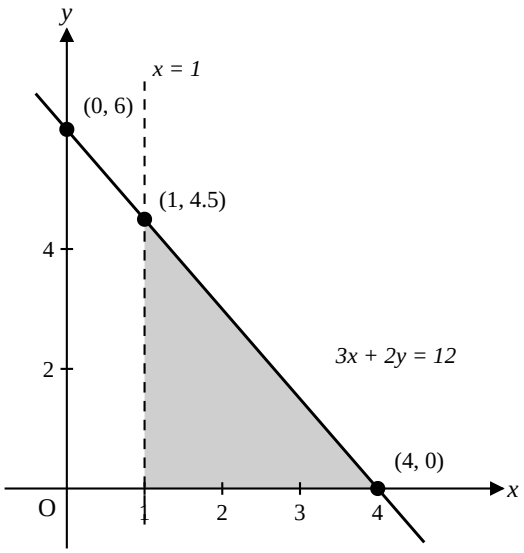
(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{6}$

(d) $\frac{\pi}{12}$

(viii) The area of the shaded region bounded by the line $3x + 2y = 12$, the line $x = 1$ and the x-axis is given by: *(Apply)* [1]



(a) $\int_1^4 (6 + \frac{3x}{2})\, dx$

(b) $\int_1^4 (6 - \frac{3x}{2})\, dx$

(c) $\int_0^6 (4 - \frac{2y}{3})\, dy$

(d) $\int_0^4 (6 - \frac{3x}{2})\, dx$

(ix) A student evaluates $\int_0^{2\pi} \sqrt{1 - \cos 2x}\, dx$ as follows. [1]

Step 1: $1 - \cos 2x = 2 \sin^2 x$, so the integrand is $\sqrt{2} \cdot \sqrt{\sin^2 x}$.

Step 2: $\sqrt{\sin^2 x} = \sin x$, so the integrand is $\sqrt{2} \sin x$.

Step 3: the integral $= [-\sqrt{2} \cos x]_0^{2\pi} = 0$.

In which step is there an error (if any)? *(Understand)*

(a) Step 1

(b) Step 2

(c) Step 3

(d) There is no error

(x) Two students evaluate $\int \sin x \cos x\, dx$. Riya writes $\frac{1}{2} \sin^2 x + C$ and Sameer writes $-\frac{1}{2} \cos^2 x + C$. Whose answer is correct? *(Analyse)* [1]

(a) Only Riya's

(b) Only Sameer's

(c) Both are correct

(d) Neither is correct

(xi) If $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} + 2\hat{k}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$ is equal to: *(Recall)* [1]

(a) 138

(b) 128

(c) −138

(d) 148

(xii) The angle between the planes $x - y + 2z = 7$ and $2x + y + z = 4$ is: *(Apply)* [1]

(a) $\pi/6$

(b) $\pi/4$

(c) $\pi/2$

(d) $\pi/3$

(xiii) $Z = ax + by$, where $a, b \geq 0$ and $a + b = 8$, has its maximum value at both the corner points (6, 9) and (9, 4) of the feasible region. The maximum value of Z is: *(Apply)* [1]

(a) 45

(b) 57

(c) 72

(d) 60

(xiv) **Statement I:** In a linear programming problem the objective function and all the constraints must be linear in the decision variables. [1]

Statement II: The coefficients of the constraints and of the objective function must be known constants that stay unchanged while the problem is studied.

(Understand)

(a) Statement I is true and Statement II is false.

(b) Statement I is false and Statement II is true.

(c) Both the statements are true.

(d) Both the statements are false.

(xv) A and B are independent events with $P(A) = 0{\cdot}3$ and $P(A \cup B') = 0{\cdot}8$. Then P(B) is: *(Analyse)* [1]

(a) 5/7

(b) 3/7

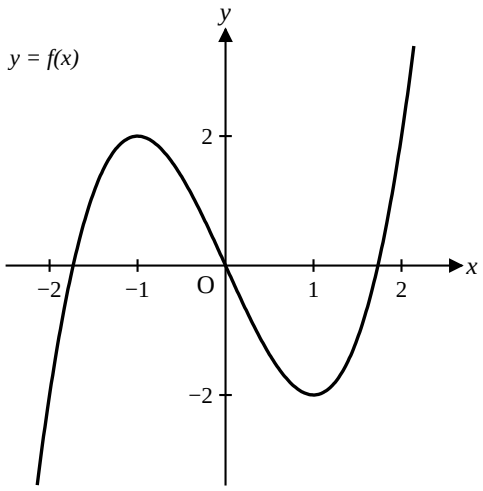
(c) 1/7

(d) 2/7

(xvi) **Assertion:** The function f : $\mathbb{R} \to \mathbb{R}$, $f(x) = x^3 - 3x$, whose graph is shown, is not one-one. [1]

Reason: If every line parallel to the y-axis cuts a curve at no more than one point, then the curve represents a function.

(Analyse)



(a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

(xvii) **Assertion:** $\int_0^{\frac{\pi}{2}} \sin x / (\sin x + \cos x)\, dx = \frac{\pi}{4}$. [1]

Reason: $\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx$ for a continuous function f.

(Understand)

(a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

(xviii) The determinant of a matrix A of order 3 is −2. If C is the matrix of cofactors of A, find the value of $|C^2|$. *(Analyse)* [1]

(xix) If m and n are the order and the degree of the differential equation $\frac{d}{dx}[(\frac{d^2y}{dx^2})^3] = 0$, find m − n. *(Understand)* [1]

(xx) The random variable X takes the values 1, 2, 3, 4, 5 with P(X = 1) = m, P(X = 2) = 2m, P(X = 3) = a, P(X = 4) = 3m and P(X = 5) = b. If P(X ≤ 2) = 0·15 and P(X ≥ 4) = 0·60, find P(X = 3). *(Apply)* [1]

SECTION B – 14 MARKS

Question 2

Find the distance from the origin of the plane that passes through the point (2, 1, 1) and is perpendicular to the line $\frac{x-1}{4} = \frac{y-3}{0} = \frac{z+2}{3}$. *(Apply)* [2]

Question 3

If $2P(A) = P(B) = \frac{7}{15}$ and $P(A \mid B) = \frac{3}{7}$, find $P(A \cup B)$. *(Apply)* [2]

Question 4

Solve the differential equation $x\,dy - y\,dx = \sqrt{x^2 + y^2}\,dx$, $x > 0$. *(Apply)* [2]

Question 5

The function $f(x) = \frac{kx}{|x|}$ for $x < 0$ and $f(x) = 4$ for $x \geq 0$ is continuous at $x = 0$. Find k. *(Understand)* [2]

OR

Find the point on the curve $y = x^2 - 6x + 5$ at which the tangent is parallel to the x-axis. *(Apply)* [2]

Question 6

Solve for x: $4\tan^{-1}x + 2\cot^{-1}x = \frac{4\pi}{3}$. *(Apply)* [2]

OR

Solve for x: $\sin^{-1}\frac{5}{x} + \sin^{-1}\frac{12}{x} = \frac{\pi}{2}$. *(Apply)* [2]

Question 7

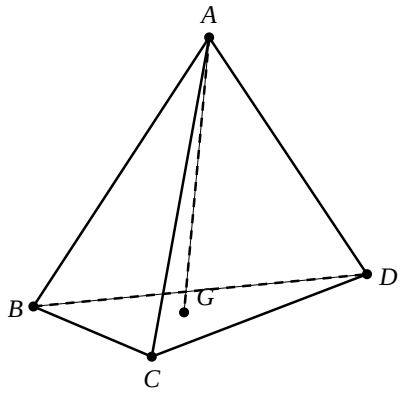
Find the unit vector perpendicular to both $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{j} + \hat{k}$. *(Apply)* [2]

OR

The vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a}| = 3$, $|\vec{b}| = \frac{2}{3}$ and $\vec{a} \times \vec{b}$ is a unit vector. Find the angle(s) between $\vec{a}$ and $\vec{b}$. *(Analyse)* [2]

Question 8

A building is to be constructed in the form of a triangular pyramid ABCD (figure, not to scale). In metres, A(0, 4, 4), B(0, 2, 0), C(1, 5, 4) and D(5, 2, 2); G is the point of intersection of the medians of ΔBCD.



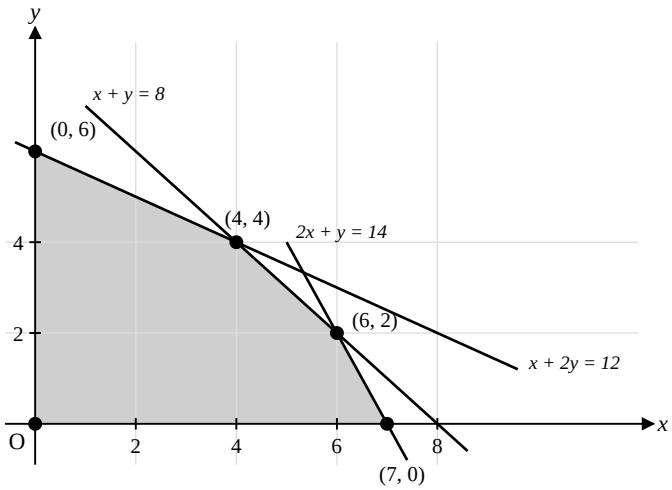
(i) Find the length of the vector $\vec{AG}$. *(Apply)* [1]

(ii) Find the area of ΔABC. *(Apply)* [1]

SECTION C – 21 MARKS

Question 9

The shaded region in the figure is the feasible region of a linear programming problem. The corner points are O(0, 0), (7, 0), (6, 2), (4, 4) and (0, 6).



(i) Write the constraints that define the feasible region. *(Understand)* [1]

(ii) Find the maximum value of Z = 5x + 7y on this region. *(Apply)* [1]

(iii) If the objective function is changed to Z = 3x + 3y, find its maximum value and the number of points at which it occurs. *(Analyse)* [1]

Question 10

If $y = (x + \sqrt{x^2 - 1})^m$, $x > 1$, prove that $(x^2 - 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - m^2y = 0$. *(Apply)* [3]

Question 11

Let $f(x) = \tan^{-1}\dfrac{\sqrt{1+x^2}-1}{x}$, $x \neq 0$.

(i) Find $f'(x)$. *(Apply)* [2]

(ii) A student claims that f has a local maximum at some point c. Use f' to evaluate the claim. *(Evaluate)* [1]

Question 12

Two planes are π_1: $x + 2y - 2z = 3$ and π_2: $2x + 4y + pz = 12$. P is the point (1, 2, 1).

(i) Verify that P lies on π_1. *(Understand)* [1]

(ii) Find p if π_2 is parallel to π_1. *(Apply)* [1]

(iii) The line through P perpendicular to π_1 meets π_2 (with this p) at Q. Find Q. *(Analyse)* [1]

OR

Find the equation of the plane containing the line $\dfrac{x+1}{3} = \dfrac{y-2}{-1} = \dfrac{3-z}{2}$ and the point (2, 0, 1). *(Apply)* [3]

Question 13

Using properties of determinants, prove that $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$. *(Analyse)* [3]

OR

A school needs decorations for 120 chairs (one bow each) and 25 tables (one cover each). Company A charges ₹14 per bow and ₹160 per cover; Company B charges ₹12 per bow and ₹180 per cover. Using matrix multiplication, find the total cost for each company and state which is cheaper. *(Analyse)* [3]

Question 14

Answer the following.

(i) Show that $\int_0^{\frac{\pi}{2}} \log \sin x \, dx = \int_0^{\frac{\pi}{2}} \log \cos x \, dx$. *(Understand)* [1]

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} (2 \log \cos x - \log \sin 2x) \, dx$. *(Apply)* [2]

Question 15

Sketch the curves $y = x^2 + 1$ and $y = x$, and the lines $x = 0$ and $x = 2$. Hence, using integration, find the area of the region bounded by them. *(Apply)* [3]

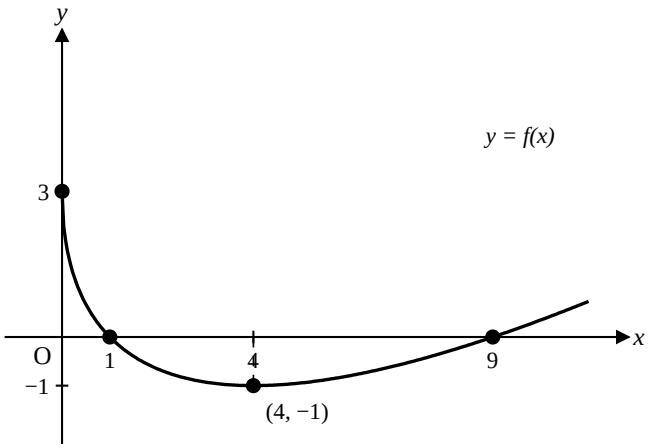
OR

The area enclosed between the curves $y = ax^2$ and $x = ay^2$, $a > 0$, is 3 square units. Find the value of a. *(Analyse)* [3]

SECTION D – 25 MARKS

Question 16

The net daily gain of a bakery, in thousands of rupees, when it bakes x hundred loaves is modelled by $f(x) = x - 4\sqrt{x} + 3$. The graph of f is shown.



(i) Write the natural domain of f. *(Recall)* [1]

(ii) Does f have an inverse on its natural domain? Justify. *(Understand)* [1]

(iii) $g : [0, 4] \to [-1, 3]$ is defined by $g(x) = f(x)$. Find $g^{-1}(x)$ and verify that $g(g^{-1}(0)) = 0$. *(Apply)* [1]

(iv) $h : [4, \infty) \to [-1, \infty)$ is defined by $h(x) = f(x)$. Find $h^{-1}(x)$ and verify that $h(h^{-1}(3)) = 3$. *(Apply)* [1]

(v) Find the value of x for which $g^{-1}(x) = h^{-1}(x)$. *(Analyse)* [1]

Question 17

A swimmer is at a point S in the sea, 3 km from the nearest point A on a straight shore. His house H is on the shore, 5 km from A. He can swim at 5 km/h and walk at 10 km/h. He swims to a point B on the shore, AB = x km ($0 \le x \le 5$), and then walks to H.

(i) Express the time T(x) taken to reach H, in hours, as a function of x. *(Understand)* [1]

(ii) Find the value of x for which the time is least. Justify that it is a minimum. *(Analyse)* [2]

(iii) Find the least time. *(Apply)* [1]

(iv) Is this less than the time he would take by swimming straight to H? Justify. *(Evaluate)* [1]

OR

A company packs a new product in a closed right circular cylindrical can of volume 250π cm^3. The base radius is r cm and the height is h cm.

(i) Find h in terms of r. *(Understand)* [1]

(ii) Find the total surface area S in terms of r. *(Apply)* [1]

(iii) Using differentiation, find the radius for which S is least. *(Analyse)* [2]

(iv) Find the minimum total surface area (in terms of π). *(Apply)* [1]

Question 18

Answer the following.

(a) Evaluate $\int \cos^{-1}x \, dx$. *(Understand)* [2]

(b) Hence evaluate $\int \tan^{-1}\sqrt{\frac{1-x}{1+x}}\,dx$. *(Evaluate)* [3]

OR

(a) Evaluate $\int x \cos x\,dx$. *(Understand)* [2]

(b) Hence evaluate $\int x^2 \sin x\,dx$. *(Evaluate)* [3]

Question 19

In a group activity class there are 10 students whose ages (in years) are 15, 16, 17, 15, 18, 17, 16, 19, 16 and 17. One student is selected at random, each student being equally likely to be chosen, and the age is recorded. Let X be the age of the selected student.

(i) Find the probability that the selected student's age is a composite number. *(Apply)* [1]

(ii) Find the probability distribution of X. *(Apply)* [2]

(iii) A student is selected and is found to be more than 15 years old. Find the probability that the age is a prime number. *(Analyse)* [1]

(iv) Find the mean age from the probability distribution. *(Apply)* [1]

OR

A medical survey finds that the chance of a patient having a heart attack is 40%. A yoga course reduces this risk by 40% of its value and a certain drug reduces it by 25% of its value. A patient chooses one of the two options with equal probabilities, and later suffers a heart attack.

(a) Define suitable events and find the probability that a randomly chosen patient suffers a heart attack. *(Apply)* [2]

(b) Find the probability that the patient had followed the yoga course. *(Analyse)* [2]

(c) Which of the two options is the more likely for a patient who had a heart attack? Is that the better option? Explain. *(Evaluate)* [1]

Question 20

Garima owns a cafe that serves burgers, samosas and juice bottles. On a particular day 160 items were sold in all. The number of burgers sold was twice the number of samosas sold, and the number of samosas sold was 20 more than the number of juice bottles sold. Form a matrix equation, solve it by the matrix method (using A^{-1}) and hence find the number of items sold in each category. *(Evaluate)* [5]

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